

MEDGREEN 2011-LB

Economic and environmental analysis for grid-connected hybrid photovoltaic-wind power system in the arid region

D.Saheb-Koussa^a, M.Koussa^a, M.Belhamel^a & M.Haddadi^b

^a Centre de Développement des Energies Renouvelable Route de l'observatoire, BP.62 Bouzareah, Alger, Algérie

^b Laboratoire de Dispositif de Communication et de Conversion Photovoltaïque E. N. P, 10 Avenue Hassen Badi, El Harrach, Alger, Algérie

Abstract

In this paper, an investigation is made on large-scale operations of 95 MW per day hybrid renewable energy system (HRES) as a grid power generation consisting of solar and wind energy. A comparison is drawn between a grid-connected HRES and a standard grid operation focusing on environmental and economic impacts. Emissions and the renewable energy generation fraction (RF) of total energy consumption are calculated as the main environmental indicator. Costs including net present cost (NPC) and cost of energy (COE) are calculated for economic evaluation. To simulate the HRES, the hourly mean global solar radiation, temperature and wind speed data from Adrar (27.59°N, 0.11°W) of Algeria, are used as an example of a typical arid climate. HOMER is used for simulation. It is found that the optimum results of HRES show a 22% reduction of emissions including CO₂, SO₂ and NO_x. The RF of the optimized system is 22%. It is also found that the reduced NPC and COE are only equal to about 99% of energy consumption from standard grid. In addition, through a set of sensitivity analysis, it is found that the wind speed has more effects on the environmental and economic performance of a HRES.

Key words: environment, economy, arid region, hybrid renewable energy system;

1. INTRODUCTION

In industrialized nations there is rapid escalation in use of fossil fuels, particularly petroleum and its by-products. In recent years increased concern about global warming, acid rain and air pollution has revitalized interest in the application of renewable energy resource [1]. Renewable energy is going to play an important role in economy due to its advantages like less emission, less waste, less energy resource use and etc. A hybrid renewable energy system making most efficient use of the different renewable resource is used to ensure stable and reliable power generation [2]. In the arid region, renewable energies have been used and are expected to improve the corresponding technical levels. [3-6]. Solar and wind energy are the main renewable energy applications. Healthcote [3] wrote that the UNESCO Arid Land Research Program introduced a 15 m diameter rotor

Nomenclature

c shape parameters (m/s)

COE cost of energy (\$/kWh)

E_{pv} energy generated by photovoltaic system (kWh)E_{oT} total energy generation (kWh)E_{WG} Energy generated by windgenerator (kWh)G global radiation(kW/m²/day)

H hub height (m)

k scale parameters

NPC net present cost (\$)

RF renewable fraction (%)

v wind speed (m/s)

z₀ surface roughness length(m)**Abbreviations**

HRES Hybrid Renewable Energy System

working in an average 20 km wind could provide 104,000 kW/ power, which would support for lighting, water heating, pumping and refrigeration for a village of 100 families. Undoubtedly, future research will discover more efficient and useful ways of using wind energy in arid region. As to solar energy, various systems have been constructed to produce electricity power by massive arrays of photovoltaic cells or reflectors focusing the solar irradiation on to the cells setting on power towers [3]. The use of solar radiation in arid lands is also seen as extensive and likely to increase in the future [4]. Some literatures investigated the efficient ways to evaluating renewable energies and reported that there is abundant amount of solar and wind energies available in different arid land. For example, Sabziparvar [5] and Sabziparvar and Shetaee [6] determined a method of calculating the global solar irradiation through comparing the different methods and taking an example of the data in Iran. It is estimated that Atacama Desert could receive in one year the equivalent of all the fossil fuels used in the world in the mid-1960s [7]. For ensuring stable and continuous power, a hybrid renewable energy system including more than one type of energy component, is often used [2]. Some researchers have studied the feasibility of using a hybrid renewable energy system in the arid region [8-11]. Mahmoudi *et al.* [12] investigated the weather data (hourly wind speed and solar radiation) for hybrid renewable energy system arid coastal countries. They assessed the feasibility of using HRES (wind + solar) in the Arabian Gulf country of Oman. However, one of the problems of HRES 's application is that there is no normal, effective and achievable method to assess the environmental and economic performance of a HRES. This study aims to discuss the environmental and economic factors of evaluating a HRES. Nevertheless, this paper analyzed the environmental and economic benefits of HRES used in the arid environment. Adrar is taken as an example of the famous Algerian the arid land. In Adrar, temperatures can vary by up to 36 °C (98.8 °F) and rainfall can vary quite dramatically from year to year. In summer, the average maximum temperature is in the high 30s, whereas in winter the average minimum temperature can be 12.6 °C (54.68 °F). The climatic data is used to simulate the long-term implementation of the system.

2 Environmental and economic assessments for HRES

Environmental and economic aspects are the two of important aspects of sustainability. To evaluate the sustainable performance of a HRES, these two impacts should be considered. The indicators of environmental and economic assessment can be determined by the five relevant meta-criteria: purpose; measurability; representativeness; reliability and feasibility; and communicability [13]. In this section, emission and renewable fraction are discussed as environmental indicators, and cost for economic indicator.

* Corresponding author. Tel.:21321901503; fax:21321901654.

E-mail address: dkoussa@cder.dz

2.1 Emission

Emission of a HRES includes carbon dioxide, sulfur dioxide and nitrogen oxides. Based on the Tokyo Protocol, CO₂ and NO_x are two types of the six main greenhouse gases [14]. SO₂ is one of the most important reasons for acid rain [15]. Emission is measured as yearly emissions of the emitted gases in kg/kWh and emissions per capital in kW/kWh. Air emission of different HRES can be estimated by the software HOMER [16]. Emission has representativeness for environmental assessment. The limitations of emission as a comprehensive measure of environmental assessment of a HRES system are from the problems associated with capturing and distinguishing all relevant negative impacts on the other aspects of environment, such as water, land and biomass diversity. In addition, emission levels do not embody information of the level of connotative impacts on long-term sustainability and health of life. However, emission is the most important reason for environmental pollution, and it is linked to the main environmental problems as greenhouse effect and acid rain. Thus, emission is used as viable headline assessment for the environmental domain. The calculated result of emission can be reliable and feasible, due to the solar radiation and wind speed data used in the simulation are reliable. They are collected from the local weather station of Adrar, Algeria. The estimating method is based on the User Manual of the software HOMER [17], and its feasibility is confirmed by many literatures mentioned before. Emission is widely accepted and understood as an environmental index. Gaseous emission has many important influences in terms of the choices, integration, and access to energy resources that make up other aspects of long-term sustainability (e.g., energy flow, material flow, and economic efficiency). In this study, the yearly emission of the hybrid renewable energy system is simulated.

2.2 Renewable fraction

Renewable fraction of a HRES means the proportion of renewable energy generated divided by total energy generated. It means the extent of renewable energy in a HRES. A greater value of this fraction presents a more renewable energy resource used. HOMER has the functions to calculate those two values and calculate the fraction directly.

Renewable fraction has representativeness for environmental assessment. It can be in turn divided into components fraction such as PV fraction (f_{pv}) and wind fraction (f_{WG}). In the study of Celik [18], the two fractions are calculated by the following equations,

$$f_{pv} = E_{pv} / E_{TOT} \quad (1)$$

$$f_{WG} = E_{WG} / E_{TOT} \quad (2)$$

where E_{pv} , E_{WG} and E_{TOT} are respectively the energy generation by photovoltaic, energy generation by wind generator and total energy generation.

2.3 Cost

The HOMER software can simulate the net present cost (NPC) and cost of energy (COE) of a hybrid energy system. The simulation input contains the initial capital, replacement cost, and operating and maintenance cost of each component of a HRES. NPC means the present value of the costs of investment and operation of a system over its lifetime. NPC is used as a main economic indicator to compare an energy system [19]. COE (\$/kWh) is the average cost per capital of useful electricity produced by the system [19]. Smaller values of NPC and COE mean a less payment to match the same electricity load. For achieving a sustainable economic efficiency, it is to minimize these two types of cost.

3 Hybrid renewable energy systems

A hybrid renewable energy system generally consists of more than one primary renewable energy component working in parallel with a secondary non-renewable component as a backup system. This study focuses on a grid- photovoltaic/wind system. Moreover, the system has a component of current converter. Fig. 1 shows a general scheme of the system. HOMER, the micropower optimization model, can simplify do the tasks of evaluating designs of grid-connected power systems for a variety of applications and energy components. Fig. 2 illustrates the proposed scheme as implemented in the HOMER code. The additional information for load, energy components, energy resources and etc, are explained in the following sections.

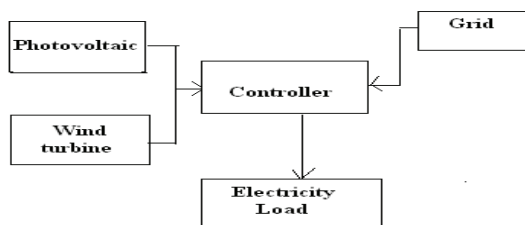


Fig. 1 Configuration of a grid-connected hybrid energy system

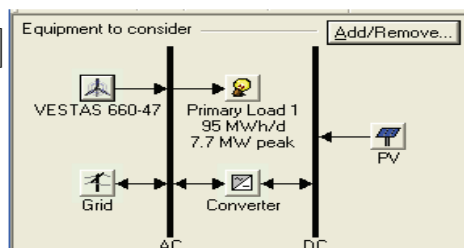


Fig. 2 Scheme of the hybrid system in HOMER code

3.1 Electrical load

Load is an important element of a HRES and any other power generating system. Air conditioner which is a main electricity consumer is used more frequently in summer. Fig. 3 shows the seasonal profile for an assumed load, an average value of 95MWh/day and a peak of 7.7 MW.

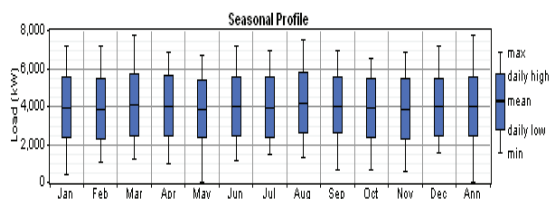


Fig. 3 Average daily load profile for a year

3.2 Renewable energy resources

3.2.1 Solar energy resources

Total daily global solar exposure derived from U.R.E.R. (MJ/m² per day) of the Adrar weather station (27.59°N,0.11°W) for the year 2005 was collected from the U.R.E.R. The data shows that the exposure duration is longer than 4516 hours per year. Scaling was done on this data to consider the long-term average annual resource (5.72 kWh/m² per day) for Adrar. HOMER introduces the clearness index from the location (latitude and longitude) information of the site under investigation. Fig. 4 demonstrates the daily radiation in kWh/m² per day and the clearness index curve over the period of the whole year. The monthly mean solar radiation is between 0.70 and 3.5 MJ/m²/day . The maximum value was seen in July

* Corresponding author. Tel.:21321901503; fax:21321901654.
E-mail address: dkoussa@cder.dz

while the minimum one was in December. As seen from the Fig.4, Adrar region has more solar resource in summer than winter. Considering the annually variations, the sensitivity analysis is done with three values around the mean, which are: 4.5, 5, 5.5, 5.93, 6.5, 7, 8, 10 kW/m² /day.

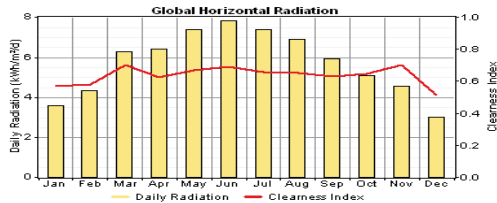


Fig. 4 Average daily global solar radiation data of Adrar

3.2.2 Wind energy resources

A hourly mean wind speed dataset (m/s) of the Adrar site's, is collected from the U.R.E.R. of Adrar, Algeria. Fig. 5 shows the monthly mean wind speed between 5.72 and 8.5m/s. Similar to solar resource, wind resource also show more affluence in summer than winter.

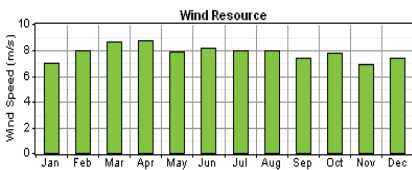


Fig. 5 Average monthly wind speed data of Adrar

The rotors of the modern wind machines are placed at heights varying between 50-100 meters, so this data was calculated at 100 meters hub height using boundary layer law [20]:

$$(U(z)/U(H)) = \log_e(z/z_0)/\log_e(H/z_0) \quad (3)$$

Where z_0 is surface roughness lengths defined in [20].

At 100 meters height, the average wind speed became 5 m/s while at 10 meters it was only 4.003 m/s. Fig. 6 demonstrates that the wind speed mainly distributes between 4 m/s and 8 m/s. As seen from Fig. 5, the wind has shortage during January, November and December. Fig. 6 demonstrates wind speed data fitting a Weibull distribution with a scale parameter $k=2.35$ and a shape parameter $c=8.81$ m/s.

To sum up, according to the data in the year 2005, renewable energy resource (including solar and wind energy) in Adrar is abundant in summer but relatively less in winter. It can be forecasted that the hybrid renewable energy system needs more grid electricity as complementarities.

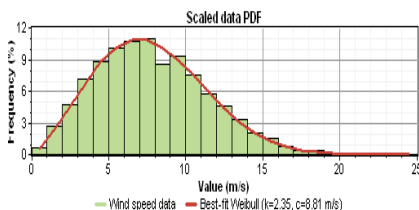


Fig. 6 Wind speed probability distribution function

3.3 Hybrid system components

The energy system components are photovoltaic modules, wind turbine, grid and power converter. This study develops a suitable assembly of the key parameters such as photovoltaic array power, wind turbine power curve, battery storage and converter capacity to match the predefined load. For economic analysis, the cost including the initial capital, replacement cost, and operating and maintenance cost are considered as simulating conditions.

3.3.1 Photovoltaic arrays

The initial cost of photovoltaic arrays may vary from \$4.00 to \$5.00 per watt [21, 22]. Considering a more optimistic system, the costs of installation, replacement and maintenance of a 1 kW solar energy system are taken as \$5000 and \$4000. Sizes of the photovoltaic arrays are varied 0, 100, 200, 300, 400, 500, 600 and 700 kW.

3.3.2 Wind turbine

Energy generation from wind turbine depends on wind speed variations. The wind turbine rated power should be greater than average electrical load. Therefore, according to the load data discussed above, the average load is around a 7.7 MW. Therefore, a VESTAS 47-660 turbine manufactured by VESTAS Windpower is used. Its rated power is 660 kW AC.

3.3.3 Grid

Grid exists as the main power component in this HRES. Moreover, grid has the functions as a storage system, so a grid power system does not need a battery [22].

3.3.4 Power converter

A converter is required for systems in which DC components serve an AC load or vice-versa. The HOMER software considers a converter as inverter (DC to AC), rectifier (AC to DC), or both. For a 1kW system the installation and replacement costs are taken as \$800 and \$750 respectively [23]. Seven different sizes of converter (0, 400, 600, 800, 1000, 1200 and 1400 kW) are taken in the model lifetime of a unit is considered to be 25 years with an efficiency of 90%.

4. Results and discussions

The software HOMER provides the results in terms of optimal systems and the sensitivity analysis. Considering the electricity price fixed at 0.4\$/kW, the PV-wind grid-connected system can be varied to identify an optimal system type for Adrar region. In this software the optimization and sensitivity results will be presented in the forthcoming paragraphs.

4.1 Optimization results

* Corresponding author. Tel.:21321901503; fax:21321901654.

E-mail address: dkoussa@cder.dz

The optimization results for specific wind speed (7.13m/s), solar irradiation (5.72 kW/m²/day) and grid electricity price (0.4\$/kWh) are summarized in Fig. 7. In this case, a wind power system seems to be most feasible economically with a minimum total net present cost (NPC) of \$177,090,600 and a minimum cost of energy (COE) of 0.379\$/kWh. This is due to the abundant wind energy resource in Adrar. In addition, the COE of wind turbine generator is more economical than solar array modules.

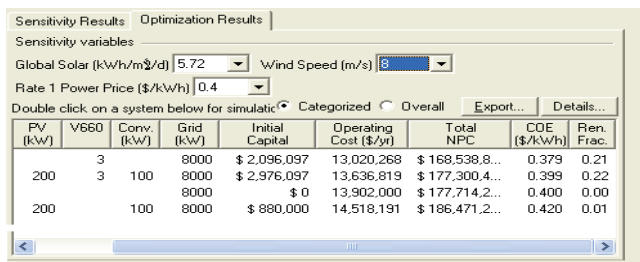


Fig. 7 The optimization results of the HOMER simulation

Fig. 7 shows that when renewable fraction was 22% the NPC was \$177,090,600 and the COE was \$0.399/kWh. This reveals that the economic performance of a PV-wind system is quite similar to the wind-grid system. The reduced NPC and COE are just equal to 99% of a standard grid power system. This PV-wind-grid case has greater renewable fraction (0.22) which means the medium proportion of renewable energy power generations. The cost of the standard grid and the optimized PV-wind-grid system, are compared in Table 1. Obviously, the PV-wind-grid system is more economical, while the NPC and COE of the standard grid system are 177,714,200\$ and 0.4\$ /kWh. Table 2 shows the emission of the standard grid system and the optimized system. As the main greenhouse gas, the emission of carbon dioxide from standard grid is 21,965,160 kW/year, while the optimized system exhausts only 17,199,524 kW/year which means 21.7 % reduction. Meanwhile, the sulphur dioxide and nitrogen oxide emissions of the PV-wind-grid system are equal to less than 78% of a standard grid system. Fig. 8 shows that the cost summary of each components of PV-wind-grid system. The most costs are for the wind turbine. In the optimized PV-wind-grid system, the grid component costs the most money (\$139,204,960), while the PV-gird system just costs \$9,090,995. This is caused by the renewable energy components do not need any more cost for energy resource, but the grid component does. The wind component and the converter cost of \$27,735,490 and \$1,065,920 respectively. The monthly energy yield of each component of the PV-wind-grid system is shown in Fig. 9. Implementing under the specific electricity load (95 MW/day), the PV array produces 398,777 kWh/year (fpv=1%). The wind turbine component produces almost 21% (7,225,602 kWh/year) of the system total energy production (34,849,388 kWh/year). In another word, the wind generation fraction f_{WG} of this system is 21%. In this system, the grid purchases share of 78% (27,225,008 kWh/year) of the total energy production.

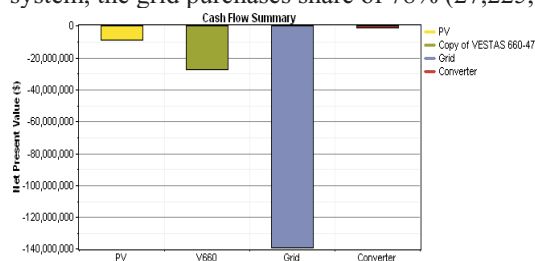


Fig. 8 Cost summary of the PV-wind-grid system

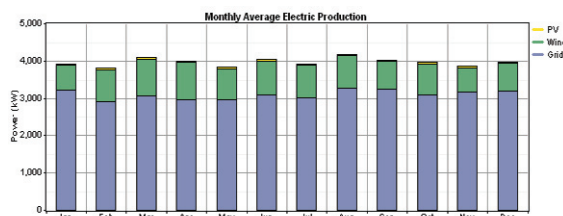


Fig. 9 Monthly average electric production of the PV- wind system

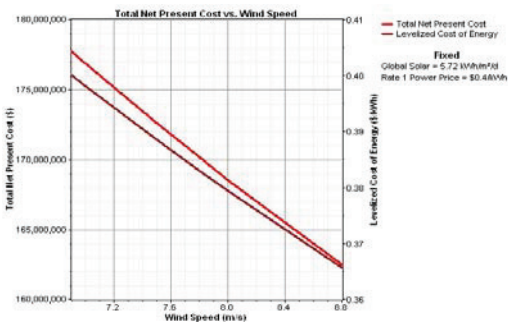


Fig. 10 The relationship between cost and wind speed

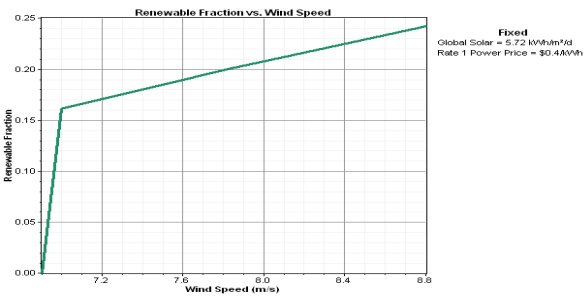


Fig. 11 The relationship between RF and wind speed

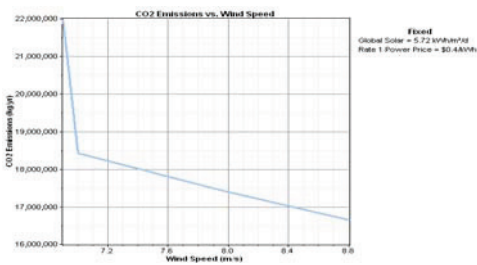


Fig. 12 The relationship between CO2 emissions and wind speed

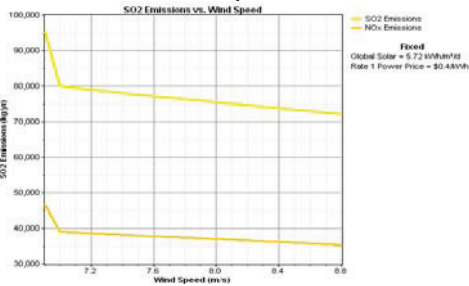


Fig. 13 The relationship between sulfur dioxide (nitrogen oxide) and wind speed

Table 1
Cost comparison between standard grid and PV-wind-grid system

Types of Costs	Standard grid	PV-wind-grid system
NPC(\$/year)	177, 714,200	177,090,600
COE(\$/kWh)	0,4	0,399

Table 2
Emissions comparison between standard grid and PV-wind-grid system

Types of Pollutant (kg/year)	Standard grid	PV-wind-grid system
CO ₂	21, 965,160	17, 225,526
NO _x	95,229	74,680
SO ₂	46,572	36,522

4.2 Sensitivity results

In this study, sensitivity analysis was done to study the effects of variation in the solar irradiation and wind speed. The simulation software simulates the long-term implementation of the hybrid system based on their respective search size for the predefined sensitivity values of the components. The emissions,

* Corresponding author. Tel.:21321901503; fax:21321901654.
E-mail address: dkoussa@cder.dz

renewable fraction, NPC and COE are simulated based on the three sensitivity variables: wind speed (m/s), solar irradiation (kW/m² /day), and grid electricity price (\$/kWh). For all of the sensitivity values HOMER simulates all the systems in their respective predefined search space. A long-term simulation for every possible system combination and configuration was done for one year period (from Jan 1st 2005 to Dec 31st 2005). In the present case, solar irradiation is set as sensitivity variables: $G=3.5, 4.5, 5.5, 5.72, 7.8, 8$ kW/m²/day, while wind speed are: $v=6.9, 7, 7.5, 7.8, 8, 8.8$ m/s . Moreover, the grid electricity price is also defined as a sensitivity variable ($p=0.1, 0.2, 0.3, 0.4$ \$/ kWh). A total of 192 sensitivity cases were tried for each system configuration. The simulation time was 23 minutes and 46 seconds on personnel computer with Intel CORE Intel Core Duo Processor of 2.53GHz and a RAM of 2 GB. The sensitivity results in terms of solar irradiation, wind speed and grid electricity price analyze the feasibility of each system. Here the feasibility of hybrid renewable energy system is analyzed based on emission reduction and cost saving. This type of sensitivity analysis of the systems provides information that a particular system would be optimal at certain sensitivity variables [24]. The PV-wind system is feasible when the grid electricity price more than \$0.4kWh/h . Under this condition, the RF can be between 0.21 and 0.22. A PV-wind system is feasible when global solar irradiation is more than 5.72kWh/m² per day and the grid electricity price is more expensive than \$0.4kWh . Based on the optimization results, wind energy production shows a bigger proportion of energy generation than solar. While the solar power occupies less than 1% of the total energy generation, wind power occupies approximately quarter. Therefore, the wind energy resource has more impacts on the implementation. Fig.s 10-13 reflects the cost, renewable fraction and emissions variation dependent on the sensitivity variable wind speed. The NPC and COE of the hybrid power system reduces when the wind speed increase from 6.9 m/s to 8.8 m/s. Simultaneously, as seen in Fig. 11, renewable fraction rise sharply from 0 to 0.16 (when wind speed increases from 6.9 to 7 m/s), and then steadily increases to 0.22 at slower rate. In addition, as shown in Fig.s 12 and 13, the main emissions of carbon dioxide, sulfur dioxide and nitrogen oxide persistently decrease 22%.

5. Conclusion

This study simulates a PV-wind grid-connected hybrid system in the arid environment. The solar and wind energy resource data are collected from the weather station of Adrar which is a typical arid region. In 2005, the mean solar irradiation was $G=5.72$ kW /m²/ day and wind speed was 7.81 m/s . The electricity load is assumed as 95 MW per day. The hybrid renewable energy system sizing is done using the software HOMER to meet the requirements of emission reduction and cost saving. The optimized HRES , including a BWC Excel-R wind turbine and an 1 kW PV module, reduces the COE to \$0.256/kWh, while the standard grid electricity is defined as \$0.3/kWh. In addition, the emissions of CO₂, SO₂ and NO_x decrease to less than 40% of the standard grid power system. The sensitivity analysis indicates that PV-wind HRES are feasible under the meteorological conditions in Adrar region. With the increasing of wind speed, the NPC, COE and emissions of the hybrid renewable energy system reduces, and renewable fraction grows up.

References

- [1] E.Bevan, *An assessment of renewable energy for the UK*. London: HMSO, Vol. 1. 1994.
- [2] D. Saheb-Koussa, , M. Haddadi, and M. Belhamel, *Economic and technical study of a hybrid system (wind-photovoltaic-diesel) for rural electrification in Algeria*. Appl Energ, 86(7-8) (2009)1024-1030.
- [3] R.L. Heathcote, *The arid lands: their use and abuse*. (1983), Hong Kong: Longman Group Limited.
- [4] F.Daniels, *Direct use of the sun's energy*. Amer. Scientist, 55(1967) 15-47.
- [5] A.A.Sabziparvar, *A simple formula for estimating global solar radiation in central arid deserts of Iran*. Renew Energ, 33(5) (2008) 1002-1010.

- [6] A.A. Sabziparvar and H. Shetaee, *Estimation of global solar radiation in arid and semi-arid climates of East and West Iran*. *Energ*, 32(5) (2007) 649-655.
- [7] E.S.Hills, *Arid lands: A geographical appraisal*. London: Mehuen & Co Ltd. ,1969
- [8] S.M. Shaahid and I. El-Amin, *Techno-economic evaluation of off-grid hybrid photovoltaic-diesel-battery power systems for rural electrification in Saudi Arabia-A way forward for sustainable development*. *Renew Sust Energ Rev* 13(3)(2009)625-633.
- [9] S.M.Shaahid and M.A. Elhadidy, *Potential of autonomous/off-grid hybrid wind-diesel power system for electrification of a remote settlement in Saudi Arabia*. *Wind Engineering*, 28(5)(2004) 621-628.
- [10] M.A. Elhadidy and S.M. Shaahid, *Feasibility of hybrid (wind+solar) power systems for Dhahran, Saudi Arabia*. *Renew Energ*, 16(1 -4) (1999) 970-976.
- [11] S.M. Shaahid and M.A. Elhadidy, *Wind and solar energy at Dhahran, Saudi Arabia*. *Renew Energ*, 4(4) (1994). 441-445.
- [12] H.Mahmoudi, *Weather data and analysis of hybrid photovoltaic-wind power generation systems adapted to a seawater greenhouse desalination unit designed for arid coastal countries*. *Desalination*, 222(1-3) (2008)119-127.
- [13] *Australia's Arid Lands*. 2005 29th, Nov 2005 [cited 2009 5th, Nov]; Available from: <http://www.cazr.csiro.au/contacts.htm>.
- [14] B.Hare, *Guide to the Kyoto Protocol*, G. International, Editor. Climate Policy Director: Amsterdam, (1998).
- [15] Y.Lang, *Acid Rain*. (2009) [cited 2009 Dec, 8]; Available from: http://en.wikipedia.org/wiki/Acid_rain.
- [16] *Using the HOMER Model in Air Quality Analysis*, P. Lilienthal and T. Lambert, Editors. National Renewable Energy Laboratory: Colorado, 2004.
- [17] *HOMER - Getting Started Guide for HOMER Version 2.1*. 2005, National Renewable Energy Laboratory, Operated for the U.S. Department of Energy Office of Energy Efficiency and Renewable Energy.
- [18] A.N.Celik, *Techno-economic analysis of autonomous PV-wind hybrid energy systems using different sizing methods*. *Energ Convers Manage*,44(12) (2003)1951-1968 .
- [19] M.Beccali, S.Brunone, M.Cellura, V.Franzitta, , *Energy, economic and environmental analysis on RET-hydrogen systems in residential buildings*. *Renew Energ*, 33(3) (2008)366-382.
- [20] T.Burton, D.Sharpe, N.Jenkins, E. Bossanyi, *Wind Energy Handbook*. West Sussex: John Wiley & Sons, Ltd,2001.
- [21] *Solar Module Price Highlights: October 2009*.(2009) [cited 2009 28th October]; Available from: <http://www.solarbuzz.com>.
- [22] J.Camerlynck, *Modelling of renewable energy system in the Maldives*,Utrecht University: Netherlands,2004.
- [23] Khan, M.J. and M.T. Iqbal, *Pre-feasibility study of stand-alone hybrid energy systems for applications in Newfoundland*. *Renew. Energ*. 30(6) (2005) 835-854.
- [24] S.Rehman, I. M.El-Amin, F. Ahmad, S. M. Shaahid, A. M. Al-Shehri, J. M. Bakhawain and A. Shash, *Feasibility study of hybrid retrofits to an isolated off-grid diesel power plant*. *Renew. Sust. Energ. Rev.*, 11(4) (2007)635-653.

* Corresponding author. Tel.:21321901503; fax:21321901654.
E-mail address: dkoussa@cder.dz